CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Vector Product

Mark scheme 13 questions · 111 marks

EXAM BOARDS AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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	Total	**111**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [***n***] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\mathbf{q} \times \mathbf{p} = (2m+3)\mathbf{i} - 5\mathbf{j} + (6-m)\mathbf{k}$	**B1**	Any form; the components need not be simplified.
	$-5 = -\lambda$, $2m+3 = \lambda$ or $6-m = \lambda$ $\Rightarrow \lambda = \ldots,\ m = \ldots$	**M1**	Equates at least two components of their $\mathbf{q} \times \mathbf{p}$ to those of $\lambda\mathbf{n}$ and solves for λ and m. Condone sign slips.
	$\lambda = 5,\ m = 1$	**A1**	
		[3]	
1(b)	$\lvert\mathbf{q} \times \mathbf{p}\rvert = \sqrt{75}$ $[= 5\sqrt{3}]$ $\lvert\mathbf{p}\rvert = \sqrt{14},\ \lvert\mathbf{q}\rvert = \sqrt{6}$	**B1**	All three, in any form.
	$\sin\theta = \frac{\text{“}5\sqrt{3}\text{”}}{\text{“}\sqrt{6}\text{”}\ \text{“}\sqrt{14}\text{”}}$ $\left[= \frac{5}{2\sqrt{7}}\right]$	**M1**	Uses $\lvert\mathbf{q} \times \mathbf{p}\rvert = \lvert\mathbf{q}\rvert\lvert\mathbf{p}\rvert \sin\theta$ with their magnitudes to find $\sin\theta$. Condone the ratio inverted.
	$\theta = 70.9°$	**A1**	awrt $70.9°$ or awrt 1.24 rad. A0 for the obtuse angle.
		[3]	
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2(a)	$\overrightarrow{AB} = \begin{pmatrix}-4\\-1\\4\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}-1\\-4\\1\end{pmatrix}$ $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\-4 & -1 & 4\\-1 & -4 & 1\end{vmatrix} = \ldots$	**M1**	Finds two sides as differences of position vectors and expands their vector product by determinant or component formula, all three components attempted. Condone sign slips; a product written without working needs two components correct.
	$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix}15\\0\\15\end{pmatrix}$	**A1**	$\pm(15\mathbf{i} + 15\mathbf{k})$, from any two sides.
	Area $= \frac{1}{2}\sqrt{\text{“15”}^2 + \text{“0”}^2 + \text{“15”}^2} = \ldots$	**dM1**	Finds half the magnitude of their vector product, using all three components.
	Area $= \frac{15\sqrt{2}}{2}$	**A1**	Accept $\frac{15}{2}\sqrt{2}$ oe; not $\frac{\sqrt{450}}{2}$, $\frac{15}{\sqrt{2}}$ or a decimal.
		[4]	
2(b)	$\mathbf{a} \times \mathbf{b} = \begin{pmatrix}8\\-12\\5\end{pmatrix}$ $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \text{“}\begin{pmatrix}8\\-12\\5\end{pmatrix}\text{”} \times \begin{pmatrix}2\\-2\\1\end{pmatrix} = \ldots$	**M1**	Expands $\mathbf{a} \times \mathbf{b}$, then the vector product of their result with $\mathbf{c}$, all three components of each attempted. Condone sign slips; each product written without working needs two components correct.
	$\mathbf{b} \times \mathbf{c} = \begin{pmatrix}9\\9\\0\end{pmatrix}$ $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \begin{pmatrix}3\\2\\0\end{pmatrix} \times \text{“}\begin{pmatrix}9\\9\\0\end{pmatrix}\text{”} = \ldots$	**M1**	Expands $\mathbf{b} \times \mathbf{c}$, then the vector product of $\mathbf{a}$ with their result, all three components of each attempted. Condone sign slips; each product written without working needs two components correct.
	$\begin{pmatrix}-2\\2\\8\end{pmatrix} \neq \begin{pmatrix}0\\0\\9\end{pmatrix}$ $\Rightarrow (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} \neq \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$	**A1 AG**	Both sides correct with no errors, and a conclusion that they differ; the two vectors joined by $\neq$ is enough.
		[3]	
	Total for Question 2	**7**	

Question	Scheme	Marks	Notes
3(a)	$\overrightarrow{AB} \times \overrightarrow{AC} = \mathbf{0}$ only if $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are parallel, or one is $\mathbf{0}$	**B1**	Accept oe such as $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \sin\theta = 0$ only when $\theta = 0$ or π, or a zero vector. The zero-vector case may be omitted.
	$\overrightarrow{AB} = \begin{pmatrix} t-1 \\ 2 \\ 4-t \end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} \neq \mathbf{0}$ y-components 2 and 0, so $\overrightarrow{AB} \neq k\overrightarrow{AC}$ for any t $\Rightarrow$ statement false for all t	**E1**	Shows $\overrightarrow{AB}$ is never a multiple of $\overrightarrow{AC}$, by components, and concludes false for all t. Not for computing the vector product.
		[2]	
3(b)(i)	$\begin{pmatrix} t-1 \\ 2 \\ 4-t \end{pmatrix} \times \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2\times 1 - (4-t)\times 0 \\ (4-t)\times 3 - (t-1)\times 1 \\ (t-1)\times 0 - 2\times 3 \end{pmatrix}$	**M1**	Forms two sides of the triangle as vectors in t and attempts their vector product, all three components. Condone sign slips.
	$= \begin{pmatrix} 2 \\ 13-4t \\ -6 \end{pmatrix}$	**A1**	Correct vector, or its negative.
	Area $= \frac{1}{2}\sqrt{2^2 + \text{“}(13-4t)\text{”}^2 + (-6)^2}$	**dM1**	Takes half the magnitude of their vector product, squaring all three components.
	$= \frac{1}{2}\sqrt{(13-4t)^2 + 40}$	**A1**	Correct expression oe, such as $\frac{1}{2}\sqrt{16t^2 - 104t + 209}$.
		[4]	
3(b)(ii)	$(13-4t)^2 + 40$ least when $13 - 4t = 0 \Rightarrow t = \ldots$	**M1**	Finds the minimum of the area or the expression under their root, by a squared term set to zero, completing the square, $t = -\frac{b}{2a}$ or a derivative set to zero.
	$t = \frac{13}{4}$	**A1**	oe, such as 3.25.
		[2]	
	Total for Question 3	**8**	

Question	Scheme	Marks	Notes
4(a)	$\overrightarrow{AB} = (p-1)\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ $\overrightarrow{AB} \cdot \mathbf{j} = \lvert\overrightarrow{AB}\rvert \cos 60°$ $\Rightarrow 2 = \frac{1}{2}\sqrt{(p-1)^2 + 2^2 + 2^2}$	**M1 A1**	**M1** Equates the scalar product of their $\overrightarrow{AB}$, a difference of position vectors, with $\mathbf{j}$ to their $\lvert\overrightarrow{AB}\rvert \cos 60°$. Condone sign slips. **A1** Any correct equation in p alone.
	$(p-1)^2 = 8 \Rightarrow p = 1 \pm 2\sqrt{2}$	**A1**	Both values, oe, or awrt 3.83 and -1.83.
		[3]	
4(b)	$\overrightarrow{OA} \times \overrightarrow{AB} = 2\mathbf{i} - 2\mathbf{j} + (3-p)\mathbf{k}$	**B1**	
	"$2\mathbf{i} - 2\mathbf{j} + (3-p)\mathbf{k}$" $= \lambda(2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ $\Rightarrow \lambda = 1,\ p = \ldots$	**M1**	Equates their vector product, with a component in p, to a multiple of the given vector, or equates ratios of components, and solves for p.
	$p = 2$	**A1**	cso.
		[3]	
4(c)	$\overrightarrow{OA} \times \overrightarrow{OB} = 2\mathbf{i} - 2\mathbf{j} + (3-p)\mathbf{k}$ $\frac{1}{2}\sqrt{2^2 + (-2)^2 + (3-p)^2} = \frac{3}{2}$	**M1 A1**	**M1** Sets half the magnitude of a vector product of two sides, in terms of p, equal to $\frac{3}{2}$. Condone slips in the vector product; needs the $\frac{1}{2}$. **A1** Any correct equation in p alone.
	$(3-p)^2 = 1 \Rightarrow p = 2$ or $p = 4$	**A1**	
		[3]	
Total for Question 4		**9**	

Question	Scheme	Marks	Notes
5(a)	$\mathbf{p} \times \mathbf{q} = \mathbf{i}(1-0) - \mathbf{j}(1-2x) + \mathbf{k}(0-x)$	**M1**	Attempts $\mathbf{p} \times \mathbf{q}$ by determinant or component formula, giving three components in terms of x. Condone sign slips.
	$= \mathbf{i} + (2x-1)\mathbf{j} - x\mathbf{k}$	**A1**	oe, as a column vector.
	$1^2 + (2x-1)^2 + (-x)^2 = 2$ $\Rightarrow 5x^2 - 4x = 0 \Rightarrow x = \ldots$	**M1**	Equates the squared magnitude of their $\mathbf{p} \times \mathbf{q}$ to 2, or its magnitude to $\sqrt{2}$, and solves for x by any means.
	$x = 0$ or $x = \frac{4}{5}$	**A1**	Both values; 0.8 accepted.
	$1 + 3(2x-1) - mx = 0$ $[\Rightarrow x(6-m) = 2]$	**M1**	Sets the scalar product of their $\mathbf{p} \times \mathbf{q}$ with $\mathbf{i} + 3\mathbf{j} + m\mathbf{k}$ equal to zero. Need not be simplified.
	$x \neq 0$, so $x = \frac{4}{5}$, $m = \frac{7}{2}$	**A1**	Both values, oe such as 0.8 and 3.5. A0 if $x = 0$ is retained as a solution.
		[6]	
5(b)	$\lvert\mathbf{p}\rvert = \sqrt{6}$, $\lvert\mathbf{q}\rvert = \frac{\sqrt{41}}{5}$	**B1**	Both, with $x = \frac{4}{5}$; oe such as $\sqrt{1.64}$.
	$\sqrt{2} = \sqrt{6} \times \frac{\sqrt{41}}{5} \sin\theta \Rightarrow \theta = \ldots$	**M1**	Uses $\lvert\mathbf{p} \times \mathbf{q}\rvert = \lvert\mathbf{p}\rvert\lvert\mathbf{q}\rvert \sin\theta$ with $\sqrt{2}$ and their magnitudes to find $\sin\theta$. Condone the ratio inverted. A scalar-product method scores M0 A0.
	$\theta = 26.8°$	**A1**	awrt 26.8° or 0.468 radians, or $\arcsin\frac{5}{\sqrt{123}}$ oe.
		[3]	
Total for Question 5		**9**	

Question	Scheme	Marks	Notes
6(a)	$\overrightarrow{OR} = 2\overrightarrow{OM} - \overrightarrow{OP} \Rightarrow R(0,1,3)$ $\overrightarrow{PR} = -2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$	**B1**	Correct R or $\overrightarrow{PR}$, soi.
	$\overrightarrow{PQ} \times \overrightarrow{PR} = (\mathbf{i} + 2\mathbf{j}) \times (-2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$	**M1**	Expands the vector product of two sides of the triangle, using their R, by determinant or component formula, all three components attempted. Condone sign slips; a product written without working needs two components correct.
	$= 4\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$	**A1**	Or its negative.
	Area $= \frac{1}{2}\sqrt{4^2 + (-2)^2 + 6^2}$	**dM1**	Takes half the magnitude of their vector product.
	$= \frac{1}{2}\sqrt{56} = \sqrt{14}$	**A1 AG**	Correct vector product and magnitude leading without error to $\sqrt{14}$.
		[5]	
6(b)	$\mathbf{n} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ $k = (2\mathbf{i} - \mathbf{j} + \mathbf{k}) \cdot (2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) = \ldots$	**M1**	Uses their vector product from (a), or a multiple, as $\mathbf{n}$, and finds its scalar product with the position vector of P, Q, R or M.
	$\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) = 8$	**A1**	Or any multiple, such as $\mathbf{r} \cdot (4\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}) = 16$. A Cartesian equation alone scores A0.
		[2]	
6(c)	dist $= \dfrac{8}{\sqrt{2^2 + (-1)^2 + 3^2}}$	**M1**	Divides $\lvert k \rvert$ by $\lvert \mathbf{n} \rvert$, both from their plane equation in (b).
	$= \frac{8}{\sqrt{14}} = \frac{4\sqrt{14}}{7}$	**A1**	Either form, or any exact equivalent.
		[2]	
	Total for Question 6	**9**	

Question	Scheme	Marks	Notes
7(a)	$\begin{pmatrix}2\\1\\20\end{pmatrix}\cdot\begin{pmatrix}10\\0\\-1\end{pmatrix}=20+0-20$ $\begin{pmatrix}2\\1\\20\end{pmatrix}\cdot\begin{pmatrix}0\\20\\-1\end{pmatrix}=0+20-20$	**M1**	Forms the scalar product of $\mathbf{n}$ with both direction vectors, or finds the vector product of the direction vectors. Condone one slip.
	Both $=0$, so $\mathbf{n}$ is normal to the field	**A1 AG**	Both products correctly evaluated as zero, or the vector product shown to be $10\mathbf{n}$, with a conclusion in words.
		[2]	
7(b)	$\mathbf{n}\times\mathbf{v_A}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&1&20\\4&-1&-2\end{vmatrix}$ $=(-2+20)\mathbf{i}-(-4-80)\mathbf{j}+(-2-4)\mathbf{k}$	**M1**	Expands $\mathbf{n}\times\mathbf{v_A}$ by determinant or component formula, all three components attempted. Condone sign slips; a product written without working needs two components correct.
	$=\begin{pmatrix}18\\84\\-6\end{pmatrix}=6\begin{pmatrix}3\\14\\-1\end{pmatrix}$, $\lambda=6$	**A1 AG**	Correct expansion leading without error to $6(3\mathbf{i}+14\mathbf{j}-\mathbf{k})$ oe, with $\lambda=6$ stated.
		[2]	
7(c)	Uses $\mathbf{v_L}$ perpendicular to $\mathbf{n}\times\mathbf{v_A}$ and to $\mathbf{n}$: $\mathbf{v_L}\cdot\begin{pmatrix}3\\14\\-1\end{pmatrix}=0,\quad\mathbf{v_L}\cdot\begin{pmatrix}2\\1\\20\end{pmatrix}=0$	**M1**	States or uses both perpendicularities, or writes $\mathbf{v_L}=\mathbf{v_A}+\mu\mathbf{n}$ with $\mathbf{v_L}\cdot\mathbf{n}=0$.
	$\begin{pmatrix}3\\14\\-1\end{pmatrix}\times\begin{pmatrix}2\\1\\20\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&14&-1\\2&1&20\end{vmatrix}=\ldots$	**dM1**	Finds a vector perpendicular to both, by a vector product or by solving the two scalar equations, or solves for μ. Condone sign slips.
	$\begin{pmatrix}281\\-62\\-25\end{pmatrix}$	**A1**	Or any positive multiple. A0 for a negative multiple. No justification of the sign is needed.
		[3]	
	Total for Question 7	**7**	

Question	Scheme	Marks	Notes
8(a)	$\mathbf{p} \times \mathbf{q} = 6\mathbf{i} - 6\mathbf{j} + 6\mathbf{k}$	**B1**	Or any non-zero multiple of $\mathbf{i} - \mathbf{j} + \mathbf{k}$; or, by scalar products, $\mathbf{r} \cdot \mathbf{p} = 2t - 2$ and $\mathbf{r} \cdot \mathbf{q} = 0$.
	$t\mathbf{i} - \mathbf{j} + \mathbf{k} = \lambda$ ("$\mathbf{i} - \mathbf{j} + \mathbf{k}$") $\Rightarrow \lambda = 1,\ t = \ldots$	**M1**	Sets $\mathbf{r}$ as a multiple of their normal vector and compares components, or sets $\mathbf{r}\times$ their normal $= \mathbf{0}$, or sets $\mathbf{r} \cdot \mathbf{p}$ and $\mathbf{r} \cdot \mathbf{q}$ equal to zero, to find t.
	$t = 1$	**A1**	
		[3]	
8(b)	$\mathbf{r} \cdot (\mathbf{p} \times \mathbf{q})$ $= (t\mathbf{i} - \mathbf{j} + \mathbf{k}) \cdot$ "$(6\mathbf{i} - 6\mathbf{j} + 6\mathbf{k})$"	**M1**	Forms the scalar product of their vector $\mathbf{p} \times \mathbf{q}$ with $\mathbf{r}$, or the determinant with rows $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ in any order.
	$= 6(t + 2)$	**A1**	$\pm 6(t + 2)$ oe, the sign depending on the order.
	$\frac{1}{6}\,\lvert$"$6(t + 2)$"$\rvert = 9 \Rightarrow t + 2 = \pm 9$ $\Rightarrow t = \ldots$	**M1**	Sets $\frac{1}{6}$ of their triple product, or its modulus, equal to 9 or ± 9 and solves for t. Needs the $\frac{1}{6}$.
	$t = 7$ or $t = -11$	**A1**	Both values and no others.
		[4]	
Total for Question 8		**7**	

Question	Scheme	Marks	Notes
9	$3\mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$ $+6\mathbf{b}\times\mathbf{a}-2\mathbf{b}\times\mathbf{b}+2\mathbf{b}\times\mathbf{c}$ $-3\mathbf{c}\times\mathbf{a}+\mathbf{c}\times\mathbf{b}-\mathbf{c}\times\mathbf{c}$	**M1**	Expands both brackets distributively into at least six products of pairs of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$. Condone sign and coefficient slips.
	$-\mathbf{a}\times\mathbf{b}-\mathbf{c}\times\mathbf{a}-6\mathbf{a}\times\mathbf{b}$ $+2\mathbf{b}\times\mathbf{c}-3\mathbf{c}\times\mathbf{a}-\mathbf{b}\times\mathbf{c}$	**M1**	Uses $\mathbf{u}\times\mathbf{u}=\mathbf{0}$ and $\mathbf{v}\times\mathbf{u}=-\mathbf{u}\times\mathbf{v}$ to reduce to multiples of the three given products. Condone one sign slip.
	$-7\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}-4\mathbf{c}\times\mathbf{a}$	**A1**	oe, such as with $+4\mathbf{a}\times\mathbf{c}$.
	$-7\begin{pmatrix}1\\2\\-1\end{pmatrix}+\begin{pmatrix}0\\3\\1\end{pmatrix}-4\begin{pmatrix}2\\-1\\4\end{pmatrix}$	**dM1**	Substitutes the given vectors into their combination of $\mathbf{a}\times\mathbf{b}$, $\mathbf{b}\times\mathbf{c}$, $\mathbf{c}\times\mathbf{a}$ and combines componentwise.
	$\begin{pmatrix}-15\\-7\\-8\end{pmatrix}$	**A1**	cao, in any vector form.
	Total for Question 9	**5**	

Question	Scheme	Marks	Notes
10(a)(i)	$(\mathbf{r}-\mathbf{q})\times(\mathbf{q}-\mathbf{p})=-2\mathbf{p}$	**B1**	
		[1]	
10(a)(ii)	$\mathbf{p}\times 2\mathbf{p}=2(\mathbf{p}\times\mathbf{p})=\ldots$	**M1**	Replaces the vector product by $2\mathbf{p}$ and uses $\mathbf{p}\times\mathbf{p}=\mathbf{0}$, or that parallel vectors have zero vector product.
	$\mathbf{0}$	**A1**	Identified as the zero vector, not the scalar 0.
		[2]	
10(a)(iii)	$\mathbf{p}\cdot 2\mathbf{p}=2\lvert\mathbf{p}\rvert^2=2(5^2)$	**M1**	Replaces the vector product by $2\mathbf{p}$ and uses $\mathbf{p}\cdot\mathbf{p}=\lvert\mathbf{p}\rvert^2$ with $\lvert\mathbf{p}\rvert=5$.
	50	**A1**	
		[2]	
10(b)(i)	OP is perpendicular to the plane PQR	**E1**	States OP is normal to plane PQR, oe such as perpendicular to both PQ and QR. Not for perpendicular to one line alone.
		[1]	
10(b)(ii)	$50\neq 0$, so O, P, Q and R are not coplanar	**E1**	States the four points are not coplanar, oe such as O not in plane PQR. Need not mention $50\neq 0$.
		[1]	
	Total for Question 10	**7**	

Question	Scheme	Marks	Notes
11(a)	$\mathbf{c} = 3\mathbf{g} - \mathbf{a} - \mathbf{b} = \ldots$	**M1**	Rearranges $\mathbf{g} = \frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c})$ for $\mathbf{c}$ and substitutes the given vectors.
	$\mathbf{c} = 5\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$	**A1**	
	$\overrightarrow{AB} = 3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\overrightarrow{AC} =$ "$3\mathbf{i} + 4\mathbf{j}$" $\overrightarrow{AB} \times \overrightarrow{AC}$ $= (0 + 12)\mathbf{i} - (0 + 9)\mathbf{j} + (12 - 6)\mathbf{k}$	**M1**	Finds $\overrightarrow{AB}$ and $\overrightarrow{AC}$, with their $\mathbf{c}$, and expands their vector product by determinant or component formula, all three components attempted. Condone sign slips; a product written without working needs two components correct.
	$= 12\mathbf{i} - 9\mathbf{j} + 6\mathbf{k}$	**A1**	Any vector form.
		[4]	
11(b)	$\mathbf{n} =$ "$4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$" $p = \mathbf{a} \cdot \mathbf{n} = 2(4) + (-1)(-3) + 3(2) = \ldots$	**M1**	Takes their (a) result, or a multiple of it, as $\mathbf{n}$ and finds its scalar product with the position vector of A, B, C or G.
	$\mathbf{r} \cdot (4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}) = 17$	**A1**	Or any multiple, such as $\mathbf{r} \cdot (12\mathbf{i} - 9\mathbf{j} + 6\mathbf{k}) = 51$. A0 for $4x - 3y + 2z = 17$ alone.
		[2]	
11(c)	$\overrightarrow{AD} = \mathbf{d} - \mathbf{a} = -\mathbf{i} + 6\mathbf{j} + \mathbf{k}$	**B1**	Or any vector from a point of Π to D, either direction, such as $\overrightarrow{BD} = -4\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$; or, by area $\times$ distance, the distance $\frac{20}{\sqrt{29}}$ from D to Π, oe.
	"$(-\mathbf{i} + 6\mathbf{j} + \mathbf{k})$" $\cdot$ "$(12\mathbf{i} - 9\mathbf{j} + 6\mathbf{k})$" $= -12 - 54 + 6$ $[= -60]$	**M1**	Forms the scalar product of their (a) result with their vector to D, or a three-edge determinant; or finds the area of ABC as $\frac{1}{2}$|their (a) result|.
	$V = \frac{1}{6}\,\lvert$"-60"$\rvert = \ldots$	**M1**	Takes one sixth of the modulus of their scalar triple product, or their $\frac{1}{3}\times$ area $\times$ distance.
	$V = 10$	**A1**	Must be positive.
		[4]	
	Total for Question 11	**10**	

Question	Scheme	Marks	Notes
12	$L_3: \ (x, y, z) = (3 + t, \ -6 - 5t, \ 1 - 2t)$ $8 + 3\lambda = 9 - 4\mu$ $3 + 2\lambda = -2 + 3\mu$ $1 - \lambda = -1 + 3\mu$	**M1**	Writes two of the lines in parametric form, with different parameters, and equates all three coordinates. Condone sign slips.
	$\Rightarrow \lambda = \ldots, \ \mu = \ldots$	**dM1**	Solves two of their equations simultaneously for both parameters, by any method.
	$\lambda = -1, \ \mu = 1 \Rightarrow L_1 \cap L_2 = (5, \ 1, \ 2)$	**A1**	One correct vertex. The vertices may be found in any order.
	$L_1 \cap L_3 = (2, \ -1, \ 3)$	**A1**	
	$L_2 \cap L_3 = (1, \ 4, \ 5)$	**A1**	
	"$(-3\mathbf{i} - 2\mathbf{j} + \mathbf{k})$" $\times$ "$(-4\mathbf{i} + 3\mathbf{j} + 3\mathbf{k})$" $= \ldots$	**M1**	Finds two sides as differences of their vertices and expands their vector product by determinant or component formula, all three components attempted. Condone sign slips; a product written without working needs two components correct.
	$= -9\mathbf{i} + 5\mathbf{j} - 17\mathbf{k}$	**A1**	Or its negative, depending on the sides chosen.
	Area $= \frac{1}{2}\sqrt{(-9)^2 + 5^2 + (-17)^2}$	**dM1**	Takes half the magnitude of their vector product, using all three components.
	$= \frac{1}{2}\sqrt{395}$	**A1**	Or awrt 9.94.
	Total for Question 12	**9**	

Question	Scheme	Marks	Notes
13(a)	$z = t \Rightarrow x = t - 2,\ y = \ldots$	**M1**	Uses both plane equations to express two coordinates in terms of a parameter, or finds a point on ℓ and the vector product of the normals.
	$\left(t-2,\ \frac{7-t}{2},\ t\right)$	**A1**	Any correct parametrisation.
	$(t-11)^2 + \left(\frac{t+13}{2}\right)^2 + (t+1)^2 = 225$	**M1**	Sets the squared distance from A to their general point equal to 225, with all three components. Condone sign slips.
	$t^2 - 6t - 27 = 0$	**A1**	Correct three-term quadratic, oe.
	$t = \ldots \Rightarrow (\ldots,\ \ldots,\ \ldots)$	**dM1**	Solves their three-term quadratic (factorising with the constants' product equal to the constant term, the correct formula with their values, or completing the square; calculator allowed) and substitutes a root into their general point.
	$B(-5, 5, -3)$, $C(7, -1, 9)$	**A1**	Both points, with B the one whose z coordinate is negative.
		[6]	
13(b)	$\overrightarrow{AB} = -14\mathbf{i} - 5\mathbf{j} - 2\mathbf{k}$ $\overrightarrow{AC} = -2\mathbf{i} - 11\mathbf{j} + 10\mathbf{k}$ $\overrightarrow{AB} \times \overrightarrow{AC} = \ldots$	**M1**	Forms two sides of triangle ABC from their points, then their vector product by determinant or component formula, all three components attempted. Condone sign slips; a product written without working needs two components correct.
	$= -72\mathbf{i} + 144\mathbf{j} + 144\mathbf{k}$	**A1**	Any non-zero multiple of $-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.
	$-x + 2y + 2z = -9 + 20 - 2$	**M1**	Forms $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$ with their normal and a point of the plane, or substitutes such a point to find the constant.
	$x - 2y - 2z + 9 = 0$	**A1**	Any Cartesian form, such as $-x + 2y + 2z = 9$; a vector equation scores A0.
		[4]	

Question 13 continued

Question	Scheme	Marks	Notes
13(c)	$\overrightarrow{AD} = -\mathbf{i} - 6\mathbf{j} + (\alpha+1)\mathbf{k}$	**B1**	Or any vector from a point of plane ABC to D, either direction; or, by area × distance, the distance $\frac{\lvert 2\alpha-9\rvert}{3}$ from D to plane ABC, oe.
	$\overrightarrow{AD} \cdot$ "$-72\mathbf{i} + 144\mathbf{j} + 144\mathbf{k}$" $= 72 - 864 + 144(\alpha+1)$ $[= 144\alpha - 648]$	**M1**	Forms the scalar product of their $\overrightarrow{AB} \times \overrightarrow{AC}$ (factor restored if rescaled) with their vector to D, or a determinant, linear in α; or finds the area of ABC as $\frac{1}{2}$|their $\overrightarrow{AB} \times \overrightarrow{AC}$|.
	$\frac{1}{6}\lvert$"$144\alpha - 648$"$\rvert = 156 \Rightarrow \alpha = \ldots$	**M1**	Sets $\frac{1}{6}$ of their triple product, or its modulus, or their $\frac{1}{3}$× area × distance, equal to 156 or ± 156 and solves for α. One value is enough.
	$\alpha = 11$ or $\alpha = -2$	**A1**	Both values and no others.
		[4]	
13(d)	Direction of ℓ: $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ $\overrightarrow{AD} = -\mathbf{i} - 6\mathbf{j} + 12\mathbf{k}$ $(2\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \times \overrightarrow{AD} = \ldots$	**M1**	Forms both lines' direction vectors, with their positive α, then their vector product by determinant or component formula, all three components attempted. Condone sign slips; a product written without working needs two components correct.
	$= -26\mathbf{j} - 13\mathbf{k}$	**A1**	Any non-zero multiple of $2\mathbf{j} + \mathbf{k}$.
	$\overrightarrow{BA} = 14\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$ $\frac{\lvert 14(0) + 5(-26) + 2(-13)\rvert}{\sqrt{26^2 + 13^2}} = \ldots$	**dM1**	Divides the scalar product of a vector joining a point of each line with their common normal by the normal's magnitude.
	$= \frac{12}{\sqrt{5}} = 5.4$	**A1**	5.4, to 2 s.f. as required; an exact or 3 s.f. answer alone scores A0.
		[4]	
	Total for Question 13	**18**	